

h) $\neg p \wedge (p \vee q) \rightarrow$ Swimming at the New Jersey shore is not allowed and either, Swimming at the New Jersey shore is allowed or Sharks have not been spotted near the shore.

12) Let p and q be the propositions "The election is decided" and "The votes have been counted," respectively. Express each of these compound propositions as an English sentence.

h) $\neg q \vee (\neg p \wedge q) \rightarrow$ Either the votes have not been counted or the election is not decided and the votes have been counted.
 a) to g) are repetitive questions done previously.

13) Let p and q be the propositions p : It is below freezing. q : It is snowing. Write these propositions using p and q and logical connectives (including negations).

- a) It is below freezing and snowing. $\rightarrow p \wedge q$.
- b) It is below freezing but not snowing. $\rightarrow p \wedge \neg q$
- c) It is not below freezing and it is not snowing. $\rightarrow \neg p \wedge \neg q$
- d) It is either snowing or below freezing (or both). $\rightarrow p \vee q$
- e) If it is below freezing, it is also snowing. $\rightarrow p \rightarrow q$
- f) Either it is below freezing or it is snowing, but it is not snowing if it is below freezing. $\rightarrow (p \vee q) \wedge (p \rightarrow \neg q)$
- g) That it is below freezing is necessary and sufficient for it to be snowing. $\rightarrow (p \leftrightarrow q)$ Book answer: $(q \leftrightarrow p) ??$

14) Let p , q and r be the propositions p : You have the flu q : You miss the final examination r : You pass the course. Express each of these propositions as an English sentence

- a) $p \rightarrow q$: If you have the flu, then you will miss the final examination.
- b) $\neg q \rightarrow \neg r$: If you will not miss the final examination, then you will pass the course.
- c) $\neg q \leftrightarrow r$: You will not miss the final examination if and only if you will pass the course.